Selective breeding: design a resilient crop

Teacher guide and worked answers

Explain selective breeding and justify a breeding plan using inherited traits, performance evidence and genetic diversity.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsl/inputs/LAUNCH KS4 - 2026-27.xlsx | LAUNCH Weekly - Spring | C30

Describe selective breeding & genetic engineering.

First of two dedicated Week 3 exemplars. Directly teaches selective breeding and its impact; genetic engineering is taught in the companion lesson. Foundation-relevant 4.8 and evaluation in 4.14; no claim to cover tissue culture.

Prior learning

Genes/alleles can be inherited; offspring from sexual reproduction vary.

A measured trait can be influenced by inherited factors and environment.

A mean summarises a set of values; larger numbers are not always better.

Success evidence

Sequence selection, reproduction, offspring testing and repeated selection.

Choose parents and offspring using both yield and damage evidence.

Explain why selected offspring still vary and why diversity matters.

Equipment

Pupil breeding record and shared candidate table; pen or mini-whiteboard.

Projector/offline deck; optional counters to mark parent choices.

Preparation

Print the shared evidence page and one response route. Table values are constructed sample data.

Read the programme rule: two parents, both disease-damage scores at most 2, at least one yield at least 45 g, different family lines where possible.

In this constructed example, yield and resistance have inherited components. No single dominant/recessive model is assigned to them.

Practical care

Paper and screen lesson; no live organisms, biological cultures or chemical work.

Ready to teach alternative

All evidence is supplied for offline use. Discuss, point, annotate or dictate instead of cutting cards or writing extended prose.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Can a breeder improve two traits at once?
2	Retrieval	2-5	Retrieve inheritance
3	I do	5-9	How selective breeding works
4	I do	9-12	Inheritance is a mechanism
5	I do	12-15	Read change across generations
6	I do	Within stage	Protect useful variation
7	We do	15-22	Breeding round 1: choose the parents
8	We do	Within stage	Breeding round 2: respond to new evidence
9	Hinge	22-25	What changes after selection?
10	You do	25-36	Write a two-generation breeding log
11	You do	Within stage	A decision needs its evidence
12	You do	Within stage	Do not promise a perfect crop
13	Review	36-39	Peer-check the breeding plan
14	Review	Within stage	A strong breeding log
15	Exit	39-40	Transfer the method

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Can a breeder improve two traits at once?

Show the outcome and predict the trade-off: 'If I choose only the biggest yield, what might I miss?' Do not introduce competition or claim pupils are plant geneticists making actual release decisions. This is a paper breeding model.

2 Retrieval Retrieve inheritance

R1 a version of a gene. R2 no; offspring inherit combinations and vary. R3 yes, for example water/light/nutrients. If R2 is insecure, point to the later sibling table before pupils make selections. Accept accurate plain-language explanations.

3 I do How selective breeding works

Think aloud through a crop goal: 'I want useful grain yield and lower disease damage. I compare candidates, select parents and breed them. I cannot order their offspring to be perfect, so I test offspring before choosing the next parents.' Describe variation as the reason selection is possible. The cycle is a model, not a one-lesson growing practical.

4 I do Inheritance is a mechanism

Use the two-locus diagram only as a compact recap. Parent AABB contributes AB and aabb contributes ab, so AaBb receives alleles from both. Do not assign dominant yield or resistance labels to these letters; those complex traits are not a monohybrid cross. Say: 'This tells me how genetic information is passed on, not the exact crop yield.'

5 I do Read change across generations

Teacher model: 'The mean increased by 10 g from generation 0 to 3 in these constructed results. I should not say every plant improved or that selection alone is proved to cause the change. The trial conditions were held the same, but a comparison line would strengthen a causal claim.' Ask a pupil to locate G2 = 43 g.

6 I do Protect useful variation

Shares the preceding 3-minute I do stage. Distinguish increased risk from certainty. Plant breeders may intentionally use inbred lines in particular programmes, but this school scenario prioritises diversity and resilience. Do not generalise family relationships to pupils or request examples from their families.

7 We do Breeding round 1: choose the parents

Use the displayed constructed table and the printed copy. Two minutes private/paired choice, then explain using both criteria. B and C meet the rule; B has damage2, C1 and C yield46. A and D fail damage despite high yield. These are choices between characterised breeding lines; the rule is deliberately strict to make an evidence-based decision possible.

8 We do Breeding round 2: respond to new evidence

Shares the preceding 7-minute We do. Reveal constructed offspring outcomes only after round1 decision. S has 50 g and damage2, so meets both traits and has the highest qualifying yield. E (45 g, damage1, unrelated F5) is a reasonable next partner to retain diversity rather than repeatedly breed close siblings. Accept P only with an explicit reason and acknowledgement of diversity risk; E is preferred under this programme. Re-test plants/offspring and use more samples before scaling up.

9 Hinge What changes after selection?

Collect independent answers. If B is not selected, return to parents → inherited alleles → varied offspring → selection. The response must say population/generations, not a plant trying harder.

A. Each plant decides to change its genes. - Individuals do not intentionally change genes to meet the breeder’s goal.

B. Desired inherited traits can become more common in later generations. - Selected parents pass on alleles, and repeated selection can change the breeding population.

C. Every offspring has exactly the same traits. - Sexual offspring vary; outcomes must be tested.

D. Growing conditions no longer matter. - Environment continues to affect crop performance.

10 You do Write a two-generation breeding log

Offer one route only. Supported = short stems/selection table; Standard = two-generation decision log; Stretch = evaluate claimed success with comparison evidence. The product should allow another scientist to understand the decision, not merely name two letters. At minute4 ask 'Which criterion ruled one out?' At minute8 check the future test.

11 You do A decision needs its evidence

Shares 11-minute independent stage. A TA may read the criteria and table aloud. Do not supply the chosen letters. Pupils may speak the log while someone records it. Stretch pupils should evaluate a control line and sampling, not simply write more.

12 You do Do not promise a perfect crop

Shares independent time. Correct 'all offspring will inherit both features' to 'some offspring may combine useful alleles, so we select after testing'. Mark explanations by content, not quantity of writing.

13 Review Peer-check the breeding plan

Light Lundy loop: choice of partner or teacher audience; pointing/speaking/writing equally welcome; teacher uses one suggestion in the shared model. Sample question: 'Where did you account for damage rather than only yield?' Give time to revise.

14 Review A strong breeding log

Shares 3-minute review. Use three formative checks; this is not a board mark scheme. Explicitly praise a changed decision justified by new evidence, not certainty or competitiveness.

15 Exit Transfer the method

E1 select parents, breed, test/compare offspring, repeat selecting suitable offspring. E2 they vary and environment affects performance. E3 reduced genetic diversity/increased chance of harmful recessive combinations or shared vulnerability. Accept precise short responses.

Answers and assessment

R1-R3 retrieval

R1 an allele is a version of a gene. R2 no: sexually produced offspring inherit combinations and can differ from both parents. R3 yes: for example light, water, nutrient availability and temperature can affect yield.

W1 / S1 / M1 first selection

B and C meet all round1 criteria: damage2 and1 (both at most 2), C mean yield46 g/plant (at least 45), and families F2/F3 differ. B’s yield40 does not disqualify it because the rule needs only one parent at or above45. Cite values with the relevant units.

S2 / M2 rejecting alternatives

A (48 g but damage6) and D (52 g but damage8) each fail the damage criterion. High yield alone may leave a crop vulnerable in the model; programme success needs both traits. Their F1 family also means A × D would not preserve different family origins.

W2 / S3 offspring decision

S has the highest yield among qualifying offspring (50 g, damage2). P also meets a45 g and damage≤2 target (45 g,1) and is defensible if prioritising lower damage; state the trade-off. Q has damage6; R has42 g, below45. Do not claim S is certain to breed true.

W3 / S4 / M3 next partner

E is preferred under the diversity goal:45 g, damage1, separate F5 family, whereas P is an F4 sibling of S. E may help retain genetic diversity but does not guarantee healthy or high-performing offspring. Re-test candidates and test many offspring in matched conditions. Accept a argued S × P plan only if the pupil explicitly explains its useful traits and the close-relative risk.

S5 sequence

Correct displayed choice order numbers: compare offspring =3; select parents =1; repeat =4; breed =2. Biological order: select desired inherited traits in parents, reproduce, compare/select offspring, repeat over generations.

S6 / E2 variation

Sexual offspring inherit different allele combinations; traits can involve many genes and environmental effects. They need testing because not all combine the intended performance, as Q's damage6 demonstrates.

M4 mechanism

Selected parents pass alleles to offspring. Offspring vary. Choosing those with useful inherited traits to reproduce again can increase those alleles/traits in the breeding population across generations. Individuals do not alter their genes because they are selected.

M5 prediction and fair test

Possible prediction: a later population may have more plants combining yield≥45 g with damage≤2. Test many offspring and a comparison population under the same light, water, soil and disease challenge, recording both traits; repeat trials. Avoid saying all offspring must improve. This is a proposed trial, not instructions to culture pathogens.

T1 follow-up calculation

Selected line:44−34=10 g/plant. Comparison:36−34=2 g/plant. Difference between changes:8 g/plant. A contemporaneous comparison helps distinguish a selection-associated shift from changes affecting both lines. It does not remove all uncertainty or prove the mechanism alone.

T2 evaluate success claim

A mean increase of10 g does not show every individual rose by10 g; individual values and spread are missing. Breeding does not guarantee outcomes. The comparison increased2 g, so factors beyond the selective programme may contribute. Replication, individual data and disease results are needed.

T3 improved evidence

One trial design improvement: larger replicated plots, randomised positions, repeat seasons/sites or retain individual values/ranges. One diversity/disease improvement: measure several disease challenges, monitor genetic diversity or include suitable unrelated lines and compare their offspring. Explain how the improvement addresses uncertainty rather than merely saying "more accurate".

T4 trade-off

Preferred S × E combines useful observed performance and distinct family origins, reducing repeated close-relative pairing. Potential trade-off: offspring still vary; E's observed yield45 is lower than S50, and a single trial may misrepresent inherited performance. S × P can be discussed with a clear close-relative risk and a testing plan.

Hinge A-D

B is correct: later generations may contain more of the selected inherited traits. A implies purposeful mutation; C ignores variation; D ignores continuing environmental influences.

E1 / E3 exit

E1 choose parents for desired inherited traits; breed them; compare and select offspring; repeat across generations. E3 accept reduced genetic diversity, greater probability of harmful recessive combinations, or shared vulnerability; do not accept that harm is guaranteed.

L1 reflection and assessment

A useful revision adds an overlooked trait value, corrects inheritance language or addresses diversity. Success requires a justified selection, accurate multigeneration mechanism and risk-aware next check. Reading/writing confidence is not the assessment target.

Teaching and assessment notes

40 minutes: opening2, retrieval3, I do10, We do7, hinge3, independent11, review3, exit1. Zero-minute slides share their stage time.

This activity is a paper model, not a practical protocol for plant infection, genetic modification or animal breeding. No live material is required.

The initial table describes breeding-line means, while the later table describes individual offspring. Teach this distinction explicitly; do not compare a line mean and an individual as if they had equal reliability.

Yield/resistance in the activity are complex traits. The AABB/aabb diagram demonstrates allele transmission only and does not assign yield/resistance to a simple dominant/recessive model.

Choose one response route. All routes preserve selection using evidence and the mechanism over generations; stretch adds evaluation of comparison data.

If pupils select D for the highest yield, cover the yield column and read the damage rule aloud, then uncover it to compare both. If pupils say all offspring improve, use Q’s damage6 as the counterexample.

A close-relative risk discussion concerns the fictitious crop families only. Do not ask pupils for personal family, health or ancestry disclosures.

Access and responsive teaching

Supported

Same scientific objective, with chunked cards, word bank, read-aloud access and short evidence stems. Accept accurately explained oral or annotated responses.

Standard

Use two evidence sources and a reasoned explanation independently.

Stretch

Evaluate limitations, explain a mechanism and transfer to unfamiliar evidence.

Regulation

Offer solo or partner working and a quiet response route. Preview the sequence; allow a brief reset and re-entry at the next numbered task. No compulsory performance or timed competition.

Misconceptions to check

The best parent produces identical perfect offspring.

Sexual offspring inherit combinations of alleles and can vary; environment also affects phenotype.

Check: Ask why sibling Q can show more damage than its parents in the constructed results.

Selective breeding changes a plant during its lifetime.

Breeders choose reproduction; inherited characteristics may become more frequent over generations.

Check: Ask which generation changes after the parent decision.

The biggest yield number is always the best choice.

A breeding objective can also require resistance, health and diversity.

Check: Compare D with B/C using both columns and the rule.

Breeding close relatives always causes disease.

It increases certain risks; it does not guarantee a harmful outcome.

Check: Ask pupils to replace “always” with a risk statement.

Word	Meaning
selective breeding	Humans choosing which organisms reproduce to increase desired inherited traits over generations.
allele	A version of a gene.
yield	Useful crop produced; here, grain mass per plant.
resistance	Ability to limit damage from a particular disease or pest.
genetic diversity	Variety of inherited genetic information in a population.
inbreeding	Breeding between close relatives, which can increase the chance of harmful recessive combinations.

Scientific references

Pearson Edexcel GCSE Biology 1BI0 specification, Topic 4

Used to check GCSE relevance; teacher-authored practice, not an official assessment. Checked 8 September 2026.

FDA: Science and History of GMOs and Other Food Modification Processes

Checked traditional selection/cross-breeding versus direct engineering. All programme numbers and family labels are original constructed sample data. Checked 8 September 2026.

RHS: F1 hybrids explained

Checked plant breeding and the purpose and drawbacks of inbred lines. Checked 8 September 2026.

Sandner et al.: Stresses affect inbreeding depression in complex ways

Primary plant research supporting risk-based, context-dependent discussion of inbreeding rather than a claim that harm is guaranteed. Checked 8 September 2026.